

AVANT TWITTER ANALYSIS

*Topic Modeling & Sentiment Analysis for Avant —
surfacing what content travels furthest across competing
fintech brands*

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Avant Twitter Analysis

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Problem Statement: The brand manager at Avant needs your help! Avant has no Twitter presence and needs strategic direction on its messaging strategy. To help get you started, she has taken the liberty of collecting tweets on a number of competing brands as well as various financial topics.

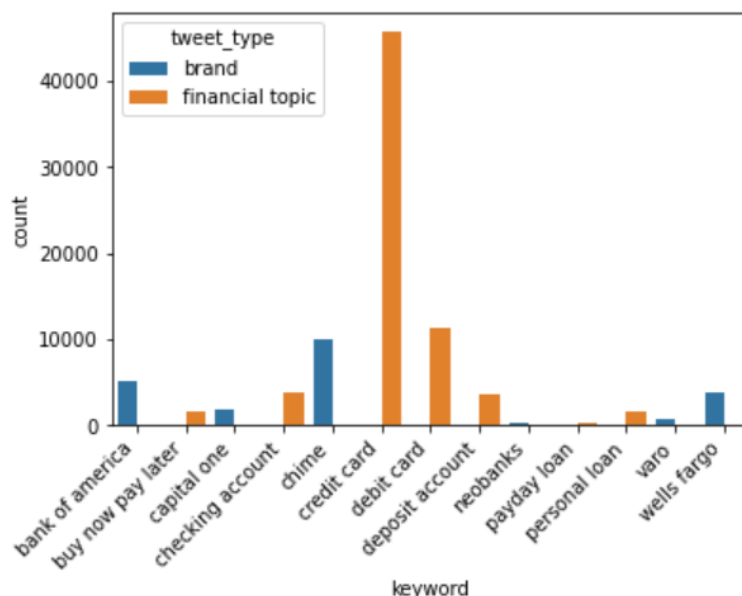
For this, I have created **topic modeling** and **sentiment analysis**.

Topic modeling is a machine learning technique that automatically analyzes text data to determine cluster words for a set of documents. I have been able to analyze some of the recurring topics across the different brands as well as some of the recurring sub-topics across the given financial categories. I have also analyzed if we have certain topics that have more traction/reach a wider audience.

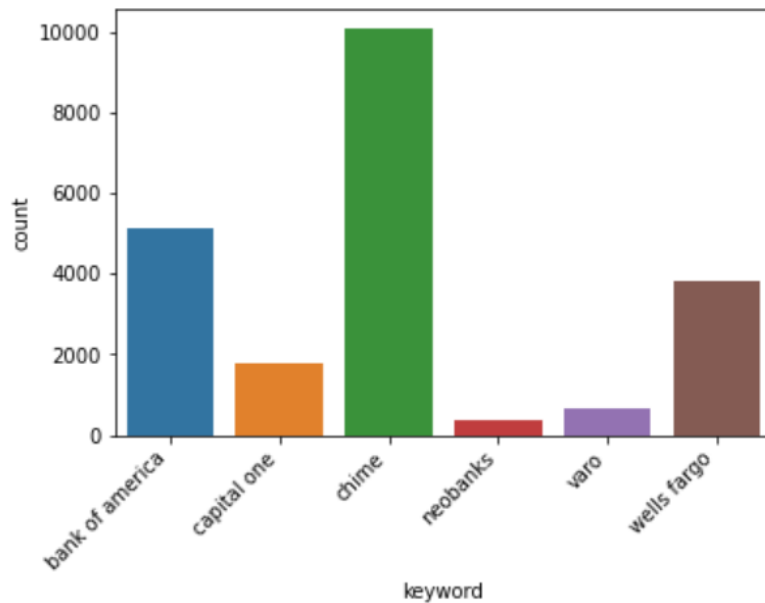
Some of the **assumptions** related to Topic Modeling are:

- Neo Bank is considered to be a brand as well as a generic term, neobanks
- Currently there are 6 brands as strong competitors to Avant: Wells Fargo, Bank of America, Capital One, Chime, Neo Bank, and Varo
- The brand names are added as stopwords so that we get unique topics listed to understand the kind of content that gets the most number of retweets
- The tweets are divided into two categories: brand terms and financial terms topics

Below is the graph that shows the most number of tweets spread across brand terms and financial topic terms. The blue bars represent the brand names and the yellow bar represents the financial topics. Amongst the brands, Chime has the most tweets associated with it, while credit card as a financial topic has the most number of tweets associated with it.



The graph below shows a deep dive of performance in terms of tweets for brands:



Following are the tweet topics for each brand:

Wells Fargo:

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Topic Names
Topic 0:
wfc profit amp target
Topic 1:
pa tickets making center
Topic 2:
chase morgan bond got
Topic 3:
center like ben account
Topic 4:
elliptic softbank analytics crypto
Topic 5:
price target price target maintains

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The image above shows the top 6 topics for Wells Fargo tweets. The topic 0 talks about the expected profit for WFC while the topic 1 is about Wells Fargo Center upcoming events' tickets in Philadelphia. Topic 2 is related to its competitor, JP Morgan Chase. Topic 3 relates to an incident back in October 2021, where Wells Fargo, Softbank invested \$60M in a crypto analytics company called, Elliptic.

Bank of America:

Topic Names
 Topic 0:
 roval nascar charlotte race
 Topic 1:
 chicago marathon chicago marathon amp
 Topic 2:
 account like ha year
 Topic 3:
 bac stadium price earnings
 Topic 4:
 earnings report beat profit
 Topic 5:
 stock minimum wage bitcoin

The image above shows the top 6 topics for Bank of America tweets. Topic 0 talks about a race sponsored by Bank of America. Topic 1 talks about the Chicago Marathon, which is a Platinum Label event while topic 2 shows that people are enquiring about the account for a year. The 4th tweet is about the quarterly earnings report of BAC, which could help people make decisions about stock investments and loans. There have been several instances with respect to change in the minimum wage by BAC and that is what the 5th topic is about.

Capital One:

Topic Names
 Topic 0:
 city year want high
 Topic 1:
 arena amp good like
 Topic 2:
 ha money account amp
 Topic 3:
 wa new time card
 Topic 4:
 day know make police
 Topic 5:
 group kync kyn exchange

The image above shows the top 6 topics for Capital One tweets. Topic 0 seems to talk about yearly high of the stock market, Topic 1 talks about the capital one arena in Washington DC. The third topic talks about a new card from Capital One or a replaced card or a stolen card. The Topic 4 is unclear although it caters to the negative sentiment due to the word 'police', which could mean theft or hacks. The last tweet talks about KYN Capital Group which is a capital finance leasing company.

Chime:

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Topic Names
Topic 0:
like time na think
Topic 1:
day wind account money
Topic 2:
theshoesurgeon chimexsrgn need theshoesurgeon chimexsrgn
Topic 3:
boost deposit send want
Topic 4:
bell app jingle gon
Topic 5:
wa good people know

```

The image above shows the top 6 topics for Chime tweets. The second tweet talks about The Shoe Surgeon which essentially is a deal between the two companies to produce shoes that are colored mint-green to make them look like credit cards, having Chime branding as well as a pocket for Chime's Credit Builder card. The 3rd tweet is all about 'Chime Spot Me Boost' where Chime allows someone to overdraw their account by \$5 extra. The 4th tweet seems irrelevant since it resonates with the Christmas bells. It looks like the last tweet talks about the good reviews about Chime.

Neo Bank:

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Topic Names
Topic 0:
crypto investment investment platforms platforms stepping
Topic 1:
pay banks wondering mobile
Topic 2:
fintech banking enricomolinari digital
Topic 3:
stealthsend xst stealth support
Topic 4:
week ai october artificialintelligence
Topic 5:
app wallet new stealthsend

```

The image above shows the top 6 topics for Neo Bank tweets. Neobanks are fintech companies that typically offer mobile-focused banking accounts with eye-catching features and Neo Bank also is a company. The first tweet is about how neobanks in general invest in cryptos while Tweet 1 is about Neo Banking on mobile. The Topic 4 talks about the artificial intelligence event called 'World's AI Week' which essentially happens in October. The last tweet is about Stealthsend so this could be another partnership.

Varo:

Topic Names
 Topic 0:
 account got card spend
 Topic 1:
 money compliance hiring risk
 Topic 2:
 wa green non plant
 Topic 3:
 remedios like going team
 Topic 4:
 need na decentraland dclwearablesbot
 Topic 5:
 varobank dreacavea joestingsusan dreacavea varobank

Topic 1 talks about a risk related money compliance which caters to the negative sentiment. The tweet of Topic 3 is irrelevant since it talks about Remedios Varo who was a Spanish painter. Topic 4 talks about Decentraland Wearables Bot where the transactions could happen via Varo.

There are certain topics related to cryptocurrency/bitcoin, some events like Chicago Marathon or WFC and some partnerships with other companies that have more traction and have reached a wider audience.

Q) What kind of sentiment surrounds the different brands? What is the relationship between sentiment and tweet reach? (Appendix)

Python sentiment analysis is a methodology for analyzing a piece of text to discover the sentiment hidden within it. It accomplishes this by combining **machine learning** and **natural language processing** (NLP). Sentiment analysis allows you to examine the feelings expressed in a piece of text.

Bank	Positive	Negative	Neutral
Bank of America	41%	19%	40%
Capital One	43%	25%	32%
Chime	43%	19%	38%
Neo Banks	35%	17%	48%
Varo	31%	17%	52%
Wells Fargo	75%	18%	7%

- Wells Fargo has the highest positive sentiment among all the tweets it is tagged in around 75%
- Chime, Bank of America and Capital One have 41% and 43% positive sentiment in all the tweets they are tagged in

- Capital One has the highest negative sentiment of around 25% in all the tweets that it is tagged in
- Varo and Neo Banks tweets are mostly neutral around 50% of all the tweets they are tagged in

Assumption: Retweet count is taken as a proxy for the popularity and reach of a particular tweet.

Approach:

The data is highly skewed with many low reach tweets around 90% of them having 0 retweets. All the tweets are bucketed into three categories based on the popularity of the tweet as measured by the retweet counts.

1. Low Reach (0 – 100 retweets)
2. Medium Reach (100 – 200 retweets)
3. High Reach (200 and above)

Across the medium level, the sentiment of the tweets is different, and positive tweets are more in number in Bank of America and Chime Banks and negative tweets are more in the Capital One category.

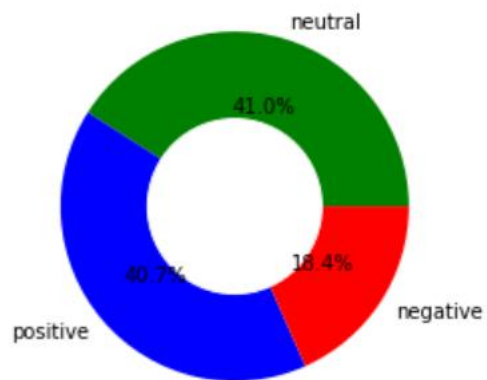
In the High Reach segment, Positive tweets are the majority in the Bank of America and the comments are mainly neutral in sentiment in Capital One and Bank of America categories.

How can Sentiment Analysis help Avant:

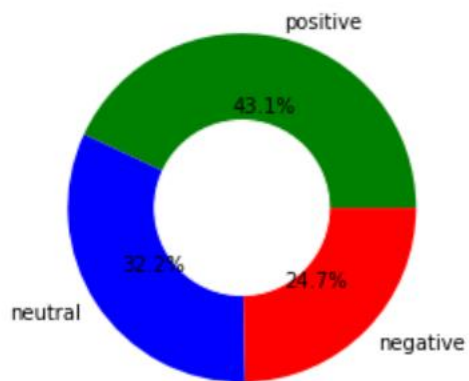
- Brand Monitoring: People use Twitter to express all sorts of feelings, hence sentiment analysis on Twitter shows you the talk about your brand and if there are negatives, then what kind of improvements can you make for those negatives and what kind of positives you can leverage further
- Social Media Monitoring: Twitter sentiment analysis allows you to keep track of what's being said about your product or service on social media, and can help you detect angry customers or negative mentions before they they escalate
- Customer Service: Because this method allows us to see positives and negatives, it helps the company to understand the critical issues first
- Market Research: Tracks specific keywords and topics to detect customer trends and interests. Understanding what customers like, what their behaviors are, and how these change over time is essential if planning to launch a new product.

Appendix:

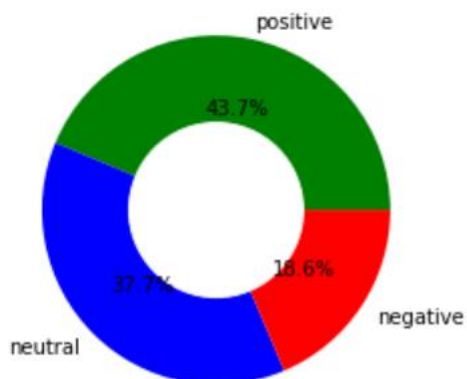
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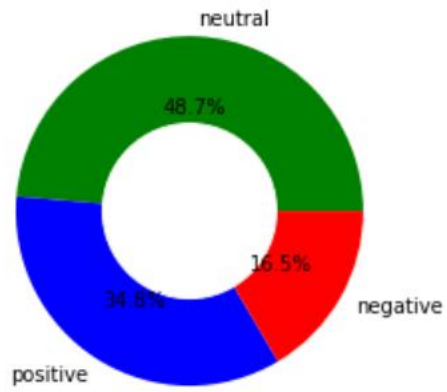
Capital One:



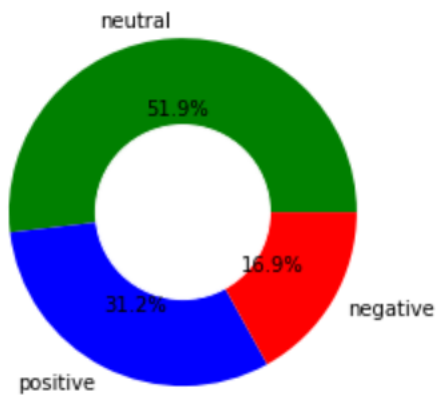
Chime:



Neo Bank:



Varo:



Wells Fargo:

